Best Books for Reverse Engineering

General reverse-engineering / fundamentals

"Practical Reverse Engineering: x86, x64, ARM, Windows Kernel, Reversing Tools, and Obfuscation" — Bruce Dang, Alexandre Gazet, Elias Bachaalany, Sebastien Josse

Level: Beginner → Intermediate

Why: Great practical intro covering architectures (x86/x64/ARM), common tools and real examples. Good bridge from theory to hands-on.

"Reverse Engineering for Beginners" — Dennis Yurichev (free ebook)

Level: Beginner

Why: Hands-on, architecture-by-architecture walkthroughs. Excellent free resource to learn core concepts and assembly.

"The IDA Pro Book: The Unofficial Guide to the World's Most Popular Disassembler" — Chris Eagle

Level: Intermediate → Advanced

Why: Deep dive into IDA Pro workflows and advanced reversing techniques. Useful even if you later use Ghidra.

Windows PE / EXE (native Windows binaries)

"Practical Malware Analysis" — Michael Sikorski & Andrew Honig

Level: Beginner → Intermediate

Why: While malware-focused, it teaches PE internals, static/dynamic analysis, and lab exercises that are directly applicable to reversing EXEs.

"Windows Internals" — Mark Russinovich, David Solomon, Alex Ionescu

Level: Intermediate → Advanced

Why: Not a RE book per se, but essential to understand Windows internals, drivers, and APIs — crucial for reversing Windows programs.

Linux / ELF

"Hacking: The Art of Exploitation" — Jon Erickson

Level: Beginner → Intermediate

Why: Strong on ELF, assembly, memory, and exploitation basics. Great for understanding how Linux binaries behave and how to analyze them.

"The Linux Programming Interface" — Michael Kerrisk

Level: Intermediate

Why: Deep system call and API reference — very helpful to understand what you're seeing in libc/ELF interactions during analysis.

Android / APK

"Android Hacker's Handbook" — Joshua J. Drake, Zach Lanier et al.

Level: Intermediate

Why: Covers Android internals, app reversing, hooking, instrumentation, and common attack surfaces.

"Mobile Application Security" — Dominic Chell, Tyrone Erasmus, Ollie Whitehouse, Mike Shema (or similar titles focused on Android reversing)

Level: Intermediate → Advanced

Why: Practical techniques for reversing obfuscated APKs, hooking with Frida, and secure coding pitfalls.

(For APKs you'll also want many hands-on blog walkthroughs and practice with tools: apktool, JADX, JADX-GUI, Frida, Objection.)

Advanced / specialist

“Practical Binary Analysis” — Dennis Andriesse

Level: Intermediate → Advanced

Why: Covers dynamic/static analysis, instrumentation, and practical binary analysis workflows.

“Binary Obfuscation and Deobfuscation” / research papers (various authors)

Level: Advanced

Why: If you plan to tackle heavily obfuscated/protected binaries (packed EXEs, obfuscated APKs), reading recent papers and tooling notes helps.

Tools & learning path (short)

Disassemblers & Decompilers: Ghidra (free), IDA (paid), Radare2 / Cutter, Binary Ninja (paid).

Android tools: apktool, JADX, ADB, Frida, Objection.

Dynamic analysis / sandbox: x64dbg / OllyDbg (Windows), GDB & pwndbg/gef (Linux), Frida for hooking on Android.

Practice labs: Crackmes, RE challenges on sites like reverse.engineering, pwnable, and CTF challenges.